Department of Education

SHS

Computer Programming (Java) (NC III)

Module 2
Java Executable Application

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What I Need To Know

This module was designed and written with you in mind. It is here to help you **understand the Integrated Development Environment (IDE)** and the pre written code or **Java Source File Structure**. The scope of this module permits it to explore the environment which enable you to easily write and debug your Java programs.

After going through this module, you are expected to:

1. Demonstrate understanding on Java Program Environment or the Integrated Development Environment (IDE)
2. Know how to download and install Java IDE (Java NetBeans)
3. Know and understand Java Source File Structure or the Application Programming Interface (API)

What I know

Directions: Choose the letter of the best answer. Use a separate sheet of paper for your answers.

1. It is a software application which enables users to more easily write and debug Java programs
   A. Integrated Development Environment
   B. Integral Development Environment
   C. Java
   D. Software Environment

2. The following are popular Java editor except.
   A. Xcode B. NetBeans C. Eclipse D. Compiler

3. Is prewritten code, organized into packages of similar topics in IDE.
   A. Java Application Programming Interface
   B. Java Virtual Machine
   C. Java Kit
   D. Java Environment

4. This is used to organize a collection of related classes.
   A. Package declaration
   B. Import Statement
   C. Class declaration
   D. Comments

5. The following are the comments included Application Programming Interface Java EXCEPT.
   A. Single Line Comment
   C. Documentation Comments

B. Block Comment D. Double comments

6. It is the fundamental component of all Java Programs.
   A. Class C. Package
   B. Comments D. Braces

7. This is used for groupings statement or block codes.
   A. Braces B. Comments C. Class D. Package

8. It is the terminating character of any java statement.
   A. Colon B. Semi colon C. Period D. Comma

9. This is the line at which the program will begin executing.
   A. Main () method C. Package
   B. Class D. Java Statement

10. This is used to reference classes and declared in other packages.
   A. Import Statement C. Class Declaration
   B. Package Statement D. Java Statement

LT 2 Integrated Development Environment (NetBeans) and Java Source File Structure

An **integrated development environment (IDE)** is a software application that provides comprehensive facilities to computer programmers for software development. An IDE normally consists of a source code editor, build automation tools, and a debugger. Most modern IDEs have intelligent code completion.

In this subject, we will use NetBeans Java editor as our Integrated Development Environment. NetBeans allows application to be developed from a set of modular software components called module. NetBeans runs on Windows, macOS, Linux and Solaris.

What's In

Directions: Write one programming environment (kung saan nagcocode ang mga programmer) you have seen or encountered. List down the words you see in that programming environment.

I. Programming Environment

_____

II. Words encountered in that Programming Environment

1. _____
2. _____
3. _____
4. _____
5. _____

What's New

Directions: Arrange the letter to form a word. (Those are Programming Environment or IDE).

1. EDOCX	
2. CEILESP	
3. SNEABTEN	
4. JBLUE	
5. NVYCODE	

What Is It

WHAT IS AN IDE?

An IDE or Integrated Development Environment enables programmer to consolidate the different aspect of writing a computer program.

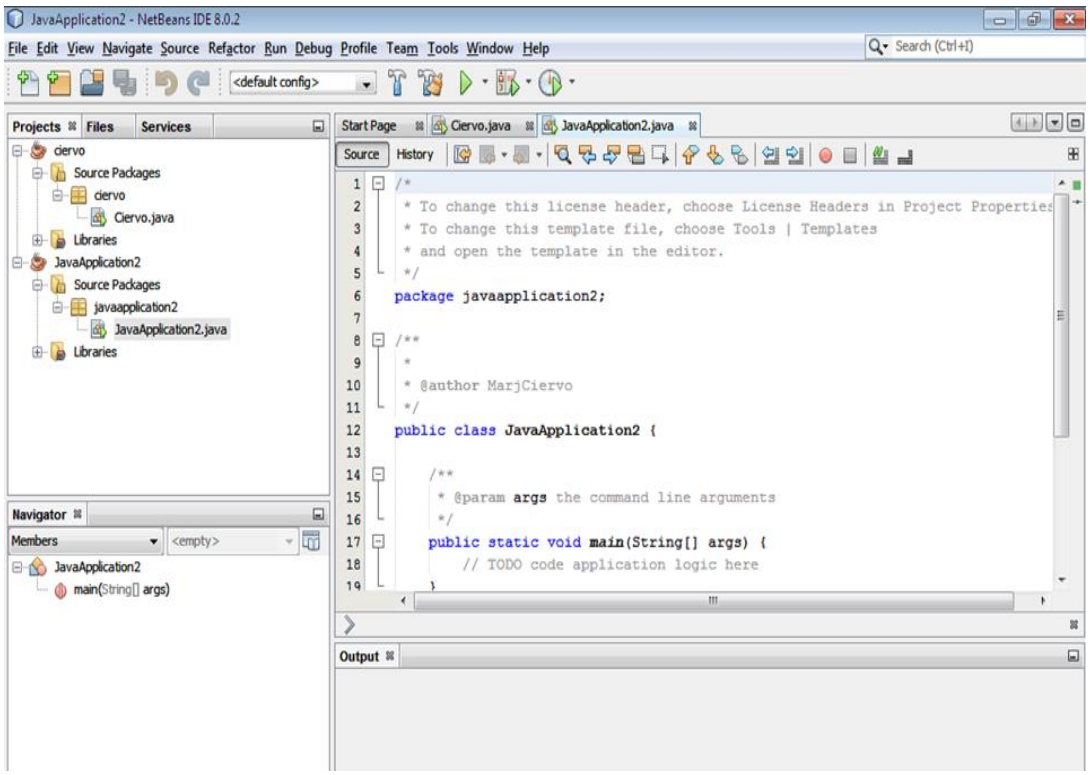


Figure 1: Image of an Integrated Development Environment (NetBeans)

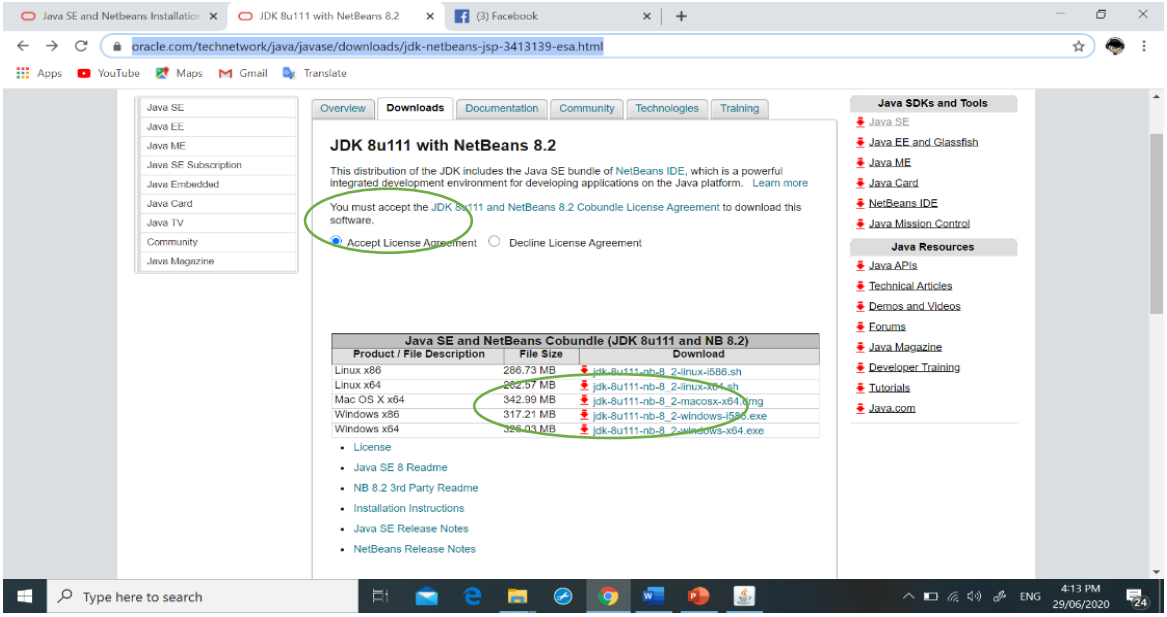
SOME OF THE BEST JAVA IDE

1. Xcode
2. Eclipse
3. **NetBeans – This is the IDE that we are going to use in this subject.**
4. BlueJ
5. Codenvy

How to download and install NetBeans

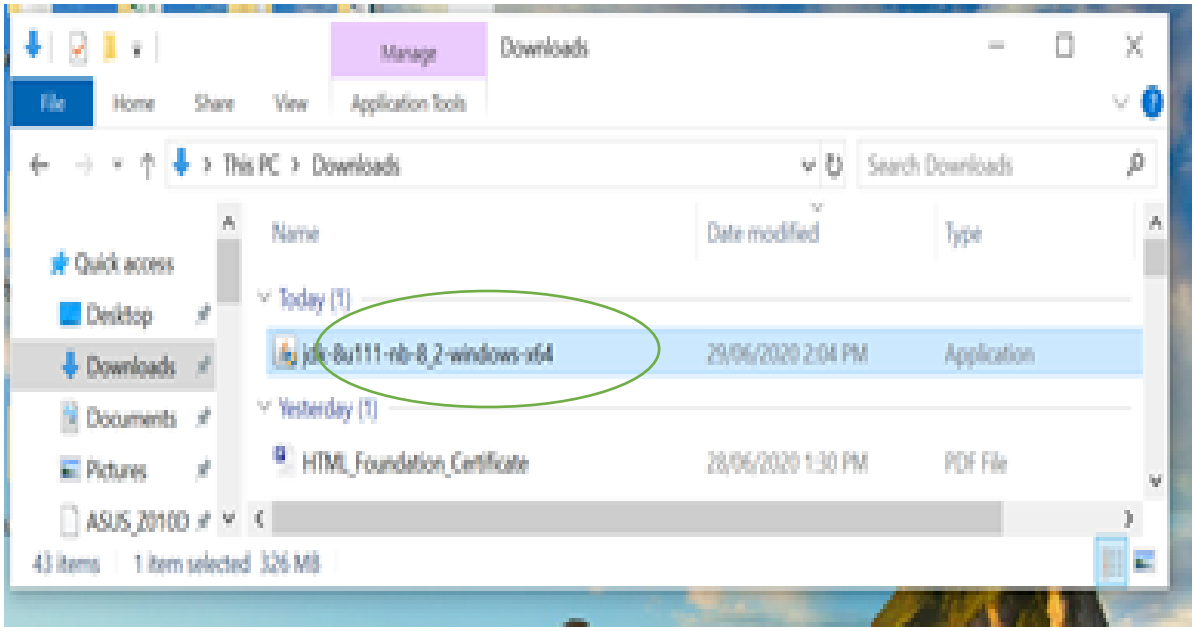
Steps on how to download NetBeans

1. Download JDK 8u111 with NetBeans 8.2 in the link: https://www.oracle.com/technetwork/java/javase/downloads/jdk-netbeans-jsp-3413139-esa.html
2. Click Accept License Agreement



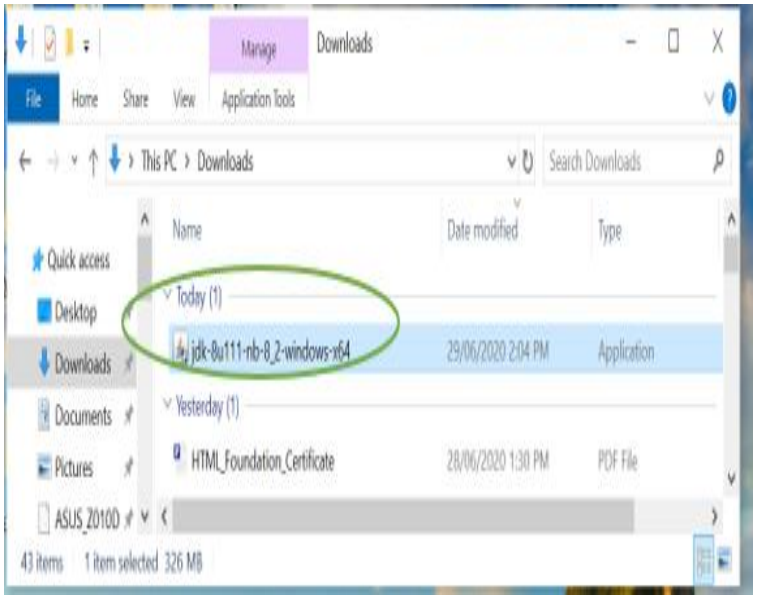
Note: It is depend on the Operating System you are using and see if you're running a 32-bit or 64-bit version of Windows.

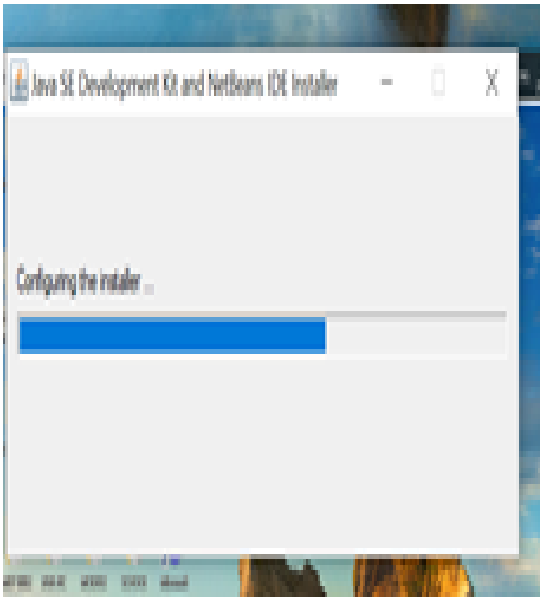
3. It automatically save in downloads.



Steps on how to install NetBeans IDE

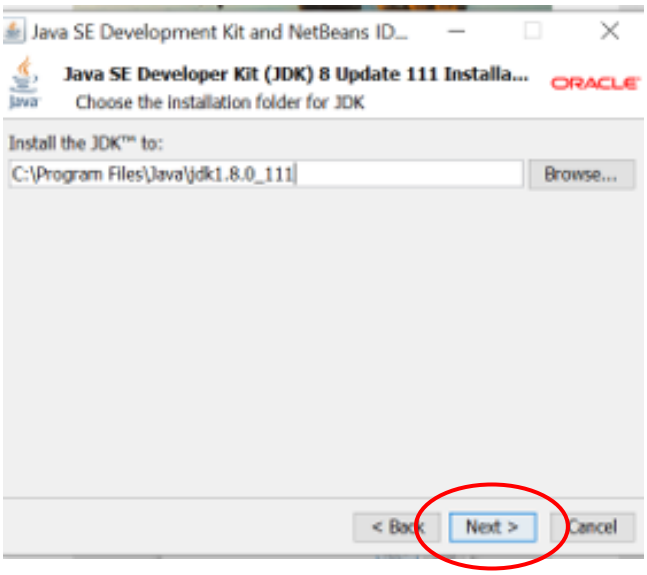
1. Double click “jdk-8u111-nb 8_2 windows-64”



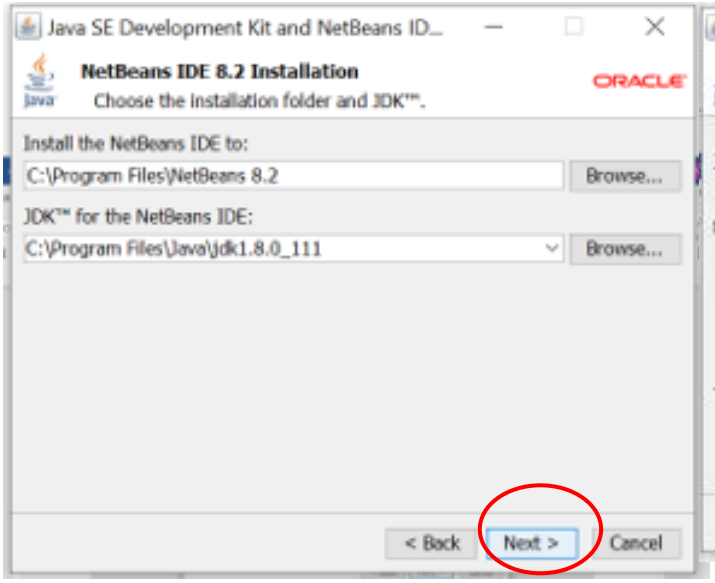


2. Then click next

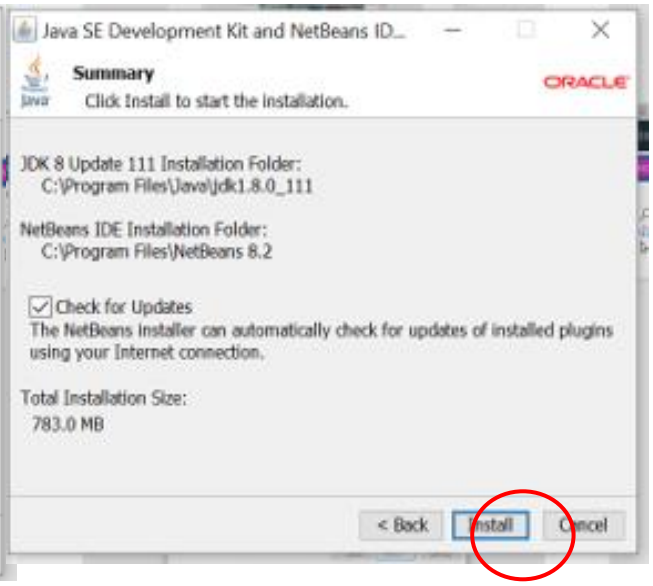
3. Click next again



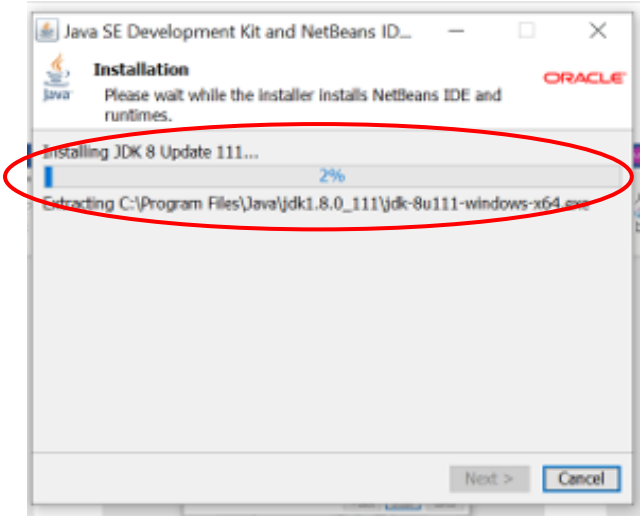
4. Click next



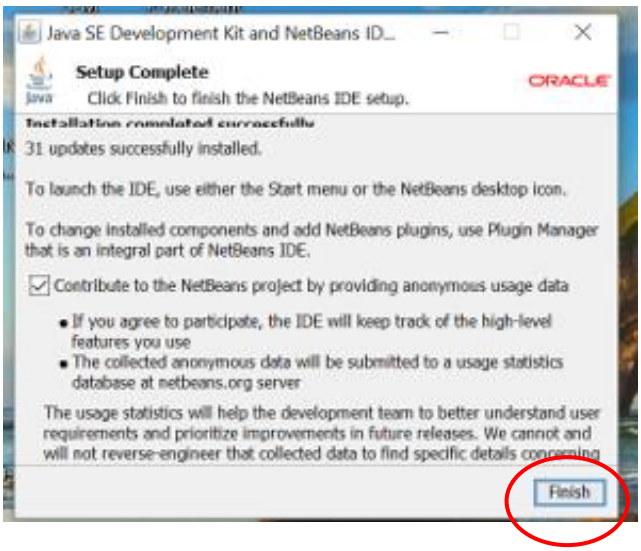
5. Click install



6. Wait for the installation to finish

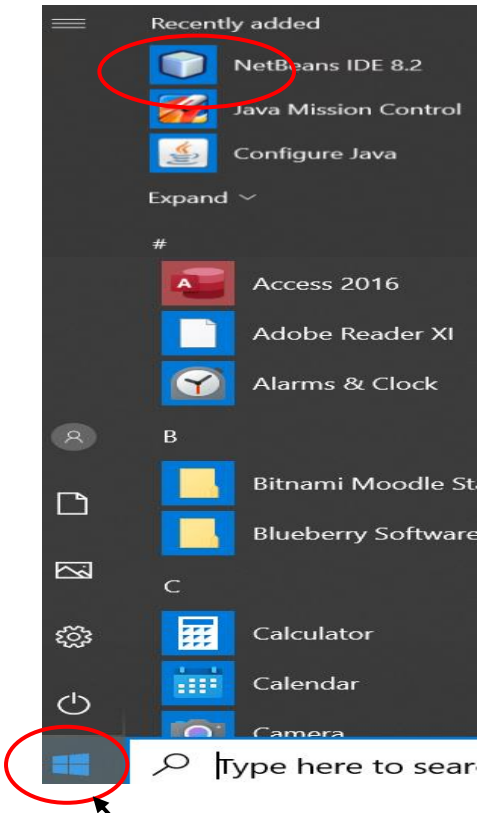


7. Click finish and your done.



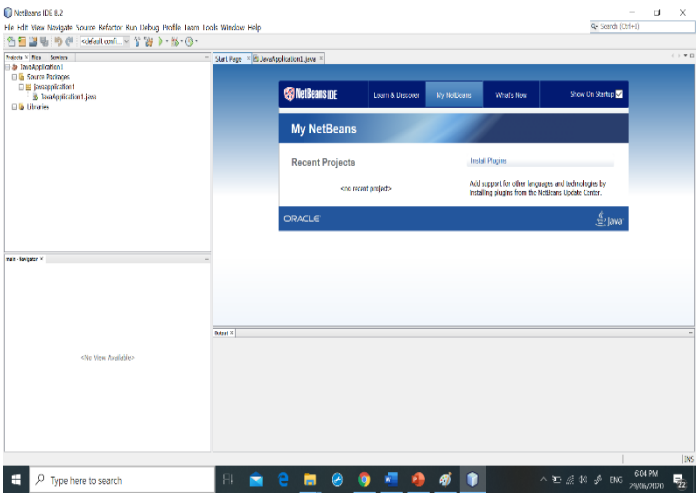
STEPS ON HOW TO OPEN NETBEANS IDE

Step 1. To start the NetBeans IDE click on **start button** then look for NetBeans IDE 8.2 or double click on the short cut icon on the desktop.

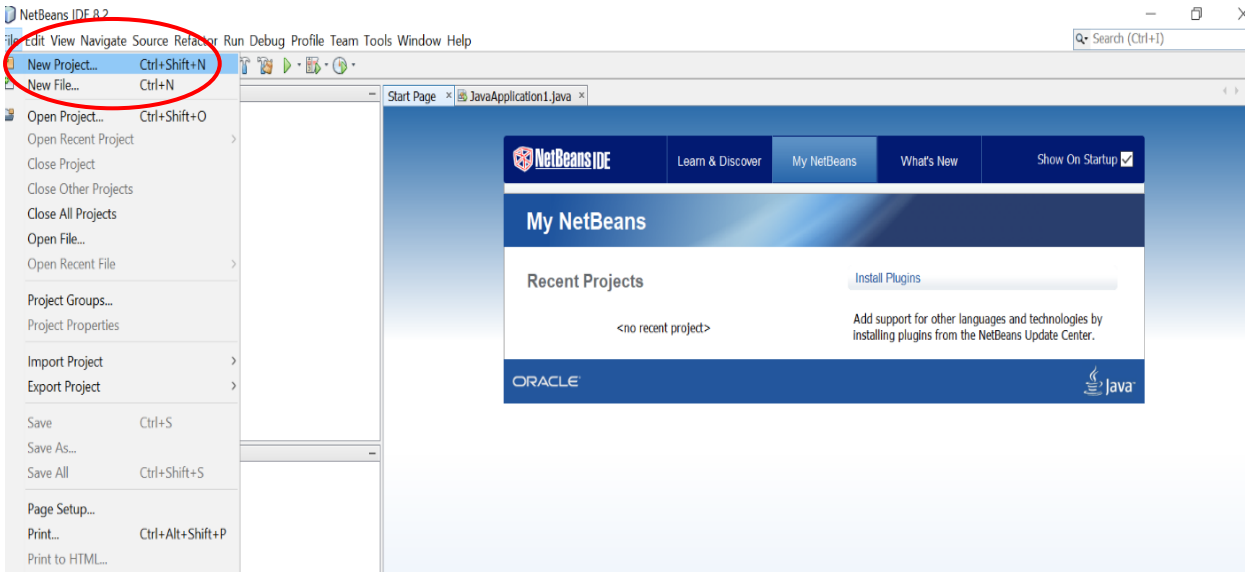


Start button

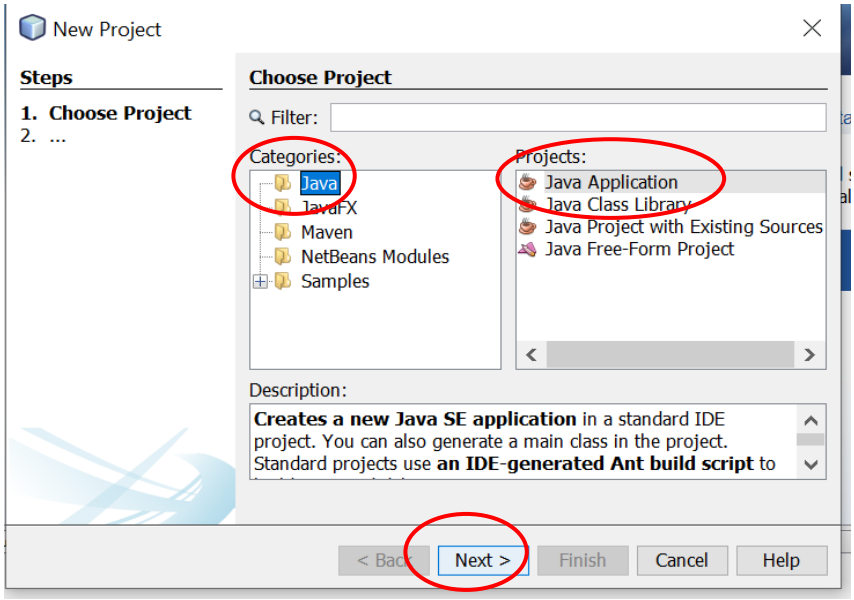
After opening NetBeans IDE, you will see the splash screen and the graphical user interface (GUI) below.



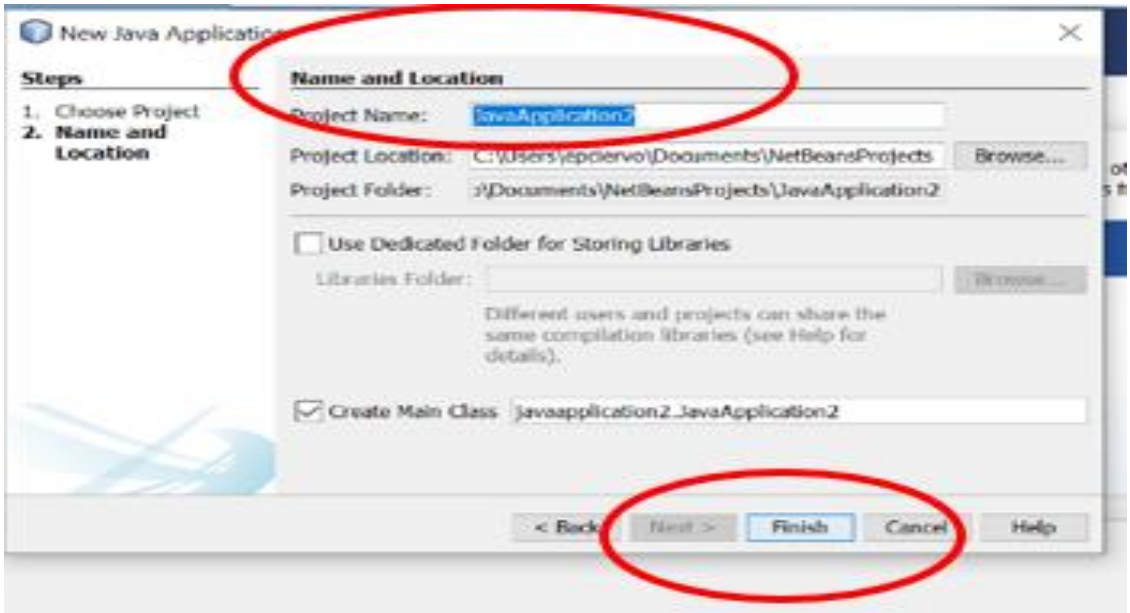
Step 2: Make a project Click on File à New Project



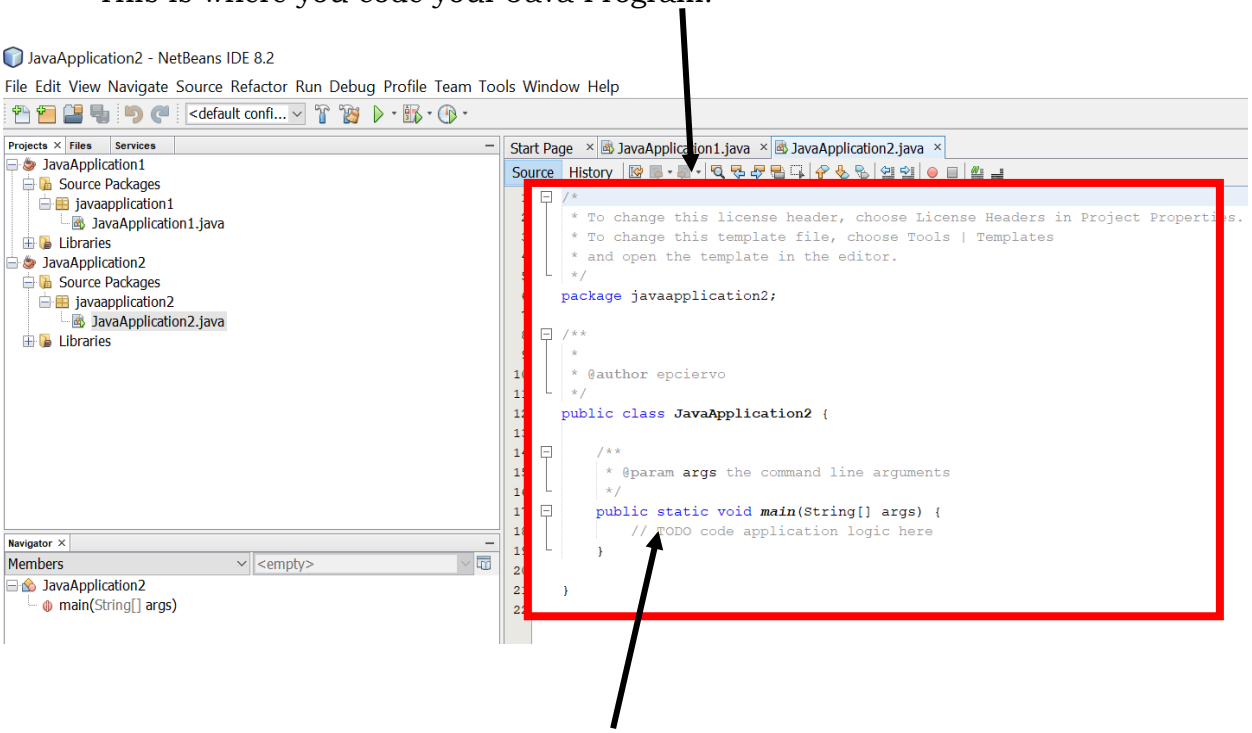
Step 3. Click Java → Java Application → Next



Step 4. Now, a New Application dialog will appear. Edit the Project Name part and type in "HelloApplication" or any file name, then click Finish.



This is where you code your Java Program.



The lesson below will explain the **pre written code** in NetBeans interface.
Note: Do not remove anything in each line except the “COMMENTS”.

JAVA SOURCE FILE STRUCTURE

DECLARATION ORDER

1. Package declaration
Used to organize a collection of classes.

2. Import statement
Used to references classes And declared in other packages

3. Classs declaration
Java source file can have sever Classes but only one public cla is allowed.

```java
/*
 *Created on July 5, 2020
 *
 * First Java Program
 */
package sampleprogram;
import java.util.*;

/**
 *
 * @author epciervo
 */
public class SampleProgram {

    public static void main(String[] args) {
        // print message here
        System.out.println("Welcome to Java Programming");
    }

}
```

COMMENTS

1. Block Comment
/*
*insert comments here
*/

2. Documentation Comment
/**
*insert
*/

3. Single Line Comment
//insert comment here

```java
/*
 *Created on July 5, 2020
 *
 * First Java Program
 */
package sampleprogram;
import java.util.*;

/**
 *
 * @author epciervo
 */
public class SampleProgram {

    public static void main(String[] args) {
        // print message here
        System.out.println("Welcome to Java Programming");
    }

}
```

White spaces
Tabs and spaces are ignored by The compiler. Use to improve Readability of code.

CLASS

- Every Java program includes at Least one class definition. The Class is the fundamental Components of all Java Programs

- A class definition contains all the variables and methods that make the program work. This is contained in the class body indicated by the opening and closing braces.

```java
/*
 *Created on July 5, 2020
 *
 * First Java Program
 */
package sampleprogram;
import java.util.*;

/**
 *
 * @author epciervo
 */
public class SampleProgram {

    public static void main(String[] args) {
        // print message here
        System.out.println("Welcome to Java Programming");
    }

}
```

Braces

- Braces are used for grouping statements or block of codes.
- The left brace ({) indicates the beginning of a class body, which contains any variables and methods the class needs.
- The left brace also indicates the beginning of a method body.
- For every left brace that opens a class or method you need a corresponding right brace (}) to close the class or method.
- A right brace always closes its nearest left brace.

```java
/*
 *Created on July 5, 2020
 *
 * First Java Program
 */
package sampleprogram;
import java.util.*;

/**
 *
 * @author epciervo
 */
public class SampleProgram {

    public static void main(String[] args) {
        // print message here
        System.out.println("Welcome to Java Programming");
    }

}
```

main () method

This line begins the main() method. This is the line at which the program will begin executing

String args[ ]

Declares a parameter named args, which is an array of String. It represents command line arguments

```
/*
 *Created on July 5, 2020
 *
 * First Java Program
 */
package sampleprogram;
import java.util.*;

/**
 *
 * @author epciervo
 */
public class SampleProgram {

    public static void main(String[] args) {
        // print message here
        System.out.println("Welcome to Java Programming");
    }

}
```

JAVA STATEMENT

- A complete unit of work in a Java program.
- A statement is always terminated with a semicolon and may span multiple lines in your source code.

System.out.println()

This line outputs the string "Welcome to Java Programmin

```
/*
 *Created on July 5, 2020
 *
 * First Java Program
 */
package sampleprogram;
import java.util.*;

/**
 *
 * @author epciervo
 */
public class SampleProgram {

    public static void main(String[] args) {
        // print message here
        System.out.println("Welcome to Java Programming");
    }

}
```

Terminating character

Semicolon(;) is the terminating character for any java statement

EXAMPLE PROGRAM

```
package sample;

public class sample {
        public static void main(String args[]) {
                System.out.println("WELCOME ICT GRADE 11");
        }
}
```

OUTPUT: **WELCOME ICT GRADE 11**

What's More

Try On

Directions: Base on your understanding explain what Integrated Development Environment is. Give at least 3 examples of IDE.

What I Have Learned

Directions: Express what you have learned in this lesson by completing the sentences below.

1. The declaration order in Java source file structure is
   a._____
   b. _____
   c. _____

2. Java Package declaration
   is_____
   _____
   _____
   _____

3. Class is
   _____
   _____
   _____
   _____

4. The comments in Java Source File structures are :
   _____
   _____
   _____
   .

   .

What I Can Do

Matching Type

Direction: Match column A with the correct answer in column B. Write only the letter of the correct answer.

COLUMN A	COLUMN B
_____1. It indicates the beginning of a class body	a. block comment
_____2. Used to organize a collection of related class	b. main method
_____3. Used to close the class or a method	c. IDE
_____4. Fundamental components of all Java Program	d. left brace
_____5. Symbol terminating character of a java statement	e.package declaration
_____6. Used to improve the readability of code`.	f. String args[]
_____7. One of the IDE	g. whitespaces
_____8. It represents command-line arguments	h. xcode
_____9 Where we code program	i. ;
_____10. Program begin to execute	j. class
	k. class

Assessment

Directions: Choose the letter of the best answer. Use a separate sheet of paper for your answers.

1. Is prewritten code, organized into packages of similar topics in IDE.
   A. Java Application Programming Interface C. Java Kit
   B. Java Virtual Machine D. Java Environment

2. The following are popular Java editor except.
   A. Xcode
   B. Netbeans
   C. Eclipse
   D. Compiler

3. This is the line at which the program will begin executing.
   A. Main () method
   B. Class
   C. Package
   D. Java Statement

4. This is used to organize a collection of related classes.
   A. Package declaration
   B. Import Statement
   C. Class declaration
   D. Comments

5. This is used for groupings statement or block codes.
   A. Braces
   B. Comments
   C. Class
   D. Package

6. It is the terminating character of any java statement.
   A. Colon B. Semi colon C. Period D. Comma

7. It software **application** which enables users to more easily write and debug **Java** programs
   A. Integrated Development Environment
   B. Integral Development Environment
   C. Java
   D. Software Environment

8. It is the fundamental component of all Java Programs.
   A. Brace
   B. Comments
   C. Package
   D. Class

9. The following are the comments included Application Programming Interface Java EXCEPT.
   A. Single Line Comment
   B. Block Comment
   C. Documentation Comments
   D. Double comments

11. This is used to reference classes and declared in other packages.
   A. Class Declaration
   B. Package Statement
   C. Import Statement
   D. Java Statement

Answer Key

1. A
2. D
3. A
4. A
5. A
6. B
7. A
8. D
9. D
10. C

References

1. https://press.rebus.community/programmingfundamentals/chapter/integrated-development-environment/#:~:text=An%20integrated%20development%20environment%20(IDE,IDEs%20have%20intelligent%20code%20completion.

2. https://netbeans.org/kb/docs/java/quickstart.html

3. https://www3.ntu.edu.sg/home/ehchua/programming/howto/netbeans_howto.html

4. https://www.oracle.com/technetwork/java/javase/downloads/jdk-netbeans-jsp-3413139-esa.html